

Tele-immersion: The Death of Distance



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It is circa 2020, a tele-mentoring session is in progress, wherein a veteran surgeon in New Delhi is step-by-step guiding a less experienced surgeon thousands of kilometres away in London. Both surgeons share the view of the surgical field and control of the robotic system and can talk using microphones. Your brain tumor has to be removed via surgery immediately. However, no senior surgeon is available in the London branch of the hospital. Tele-immersion allows dissection without the need of the senior surgeon even touching the tumor. The operation is successful and your life is saved.

A few weeks later, you schedule a very important meeting with your business associates in Bengaluru. However, you have visitors coming in from New York to sign a mega business deal the same day. Without any stress, you go to a room called holodeck. There, inside a simulated environment, using information technology you contact your business associates. You are able to conduct the meeting with them almost as if you

are in Bengaluru. You even manage to shake hands with holographic images of your business associates. With a smile on your face you proceed to receive your guests from New York.

What is tele-immersion all about

Electronic visualisation laboratory (EVL), University of Illinois, defines tele-immersion as the union of networked virtual reality and video in the context of significant computing and data mining.

Tele-immersion is a technology to be implemented with the Internet2 that will enable users in different geographic locations to come together in a simulated holographic environment to interact. To users, it would appear as if they are actually interacting, communicating and meeting each other in person at a common location. In a tele-immersive environment, computers recognise the presence and movements of individuals as well as physical and virtual objects. These can then track living and non-living objects and project them in a realistic manner across many geographic locations. Holographic environments are

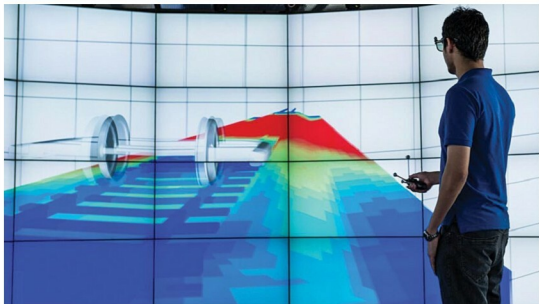


Fig. 1: Virtual train tracks as seen in EVL's CAVE2 (Source: www.evl.uic.edu)